METRO TICKET GENERATING SYSTEM

Project Design Phase-II – Technology Stack

Date	**2 October 2026**
Team ID	
Project Name	**Metro Ticket Generating System in ServiceNow Using Service Catalog and Workflows**
Maximum Marks	**4 Marks**

Reference: IBM Cloud Architecture Patterns (https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/)

Technical Architecture

The proposed architecture is a ServiceNow cloud-platform custom-scoped workflow architecture (x_metro_ticket). The commuter interacts with the Service Catalog / Service Portal and custom portal QR widgets, the request is represented by a Requested Item record and custom scoped booking tables, and Flow Designer / Workflow Studio automates validation, dynamic script calculations, QR code payload generation, operator task creation, and email notifications.

Table-1: Components & Technologies

- **1. User Interface:** How the commuter interacts with the metro ticket request and digital pass view | *Technology:* ServiceNow Service Portal / Service Catalog / Custom QR Portal Widget
- **2. Application Logic-1:** Captures the request and manages catalog-request processing via custom tables | *Technology:* ServiceNow Custom Scoped App (x_metro_ticket) / Record Producer
- **3. Application Logic-2:** Automates dynamic fare calculations, script execution, and condition-based flow execution | *Technology:* ServiceNow Script Includes & Flow Designer / Workflow Studio
- **4. Application Logic-3:** Creates the operator task and generates the cryptographic QR code payload after validation | *Technology:* ServiceNow Catalog Task Designer & Flow Designer Action
- **5. Database:** Stores catalog requests, Requested Items, custom station tables, booking records, and operator tasks | *Technology:* ServiceNow platform database (Custom Scoped Tables)
- **6. Cloud Database:** Cloud-hosted application data storage provided by the ServiceNow platform | *Technology:* ServiceNow cloud platform
- **7. File Storage:** No separate file-storage service is required for the core workflow | *Technology:* Not applicable
- **8. External API-1:** No external API is required by the implemented core workflow | *Technology:* Not applicable
- **9. External API-2:** No second external API is required by the implemented workflow | *Technology:* Not applicable
- **10. Machine Learning Model:** No machine-learning model is used in this project | *Technology:* Not applicable
- **11. Infrastructure (Server / Cloud):** Application and workflow run on the ServiceNow developer instance | *Technology:* ServiceNow cloud platform

Table-2: Application Characteristics

- **1. Open-Source Frameworks:** No separate open-source application framework is required for the core implementation | *Technology:* Not applicable; ServiceNow platform
- **2. Security Implementations:** Access to records, custom tables, and configuration is controlled through ServiceNow authentication, scoped roles, and Access Control Lists (ACLs) | *Technology:* ServiceNow roles / Scoped ACL-based access control

- **3. Scalable Architecture:** The solution separates request capture, dynamic script calculation, QR generation, and operator fulfillment automation into platform components that can be extended | *Technology:* ServiceNow Service Catalog + Flow Designer + Scoped App Studio
- **4. Availability:** The application is accessed through the ServiceNow cloud instance rather than a locally hosted application server | *Technology:* ServiceNow cloud platform
- **5. Performance:** The flow uses event-driven automation: a submitted catalog request triggers instant server-side script calculation, approval routing, QR payload generation, and operator task assignment without manual processing steps | *Technology:* ServiceNow Flow Designer & Server-Side Scripts